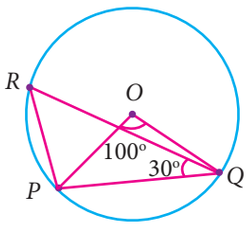


standard students for planting sapplings. Four students plant trees at the points A, B, C and D as shown in figure. Here $AB = 8$m, $CD = 10$m and $AB \perp CD$. If another student places a flower pot at the point P, the intersection of AB and CD, then find the distance from the centre to P.

9. In the given figure, $\angle POQ = 100^0$ and $\angle PQR = 30^0$, then find $\angle RPO$.



4.7 Practical Geometry

Practical geometry is the method of applying the rules of geometry dealt with the properties of points, lines and other figures to construct geometrical figures. "Construction" in Geometry means to draw shapes, angles or lines accurately. The geometric constructions have been discussed in detail in Euclid's book 'Elements'. Hence these constructions are also known as Euclidean constructions. These constructions use only compass and straightedge (i.e. ruler). The compass establishes equidistance and the straightedge establishes collinearity. All geometric constructions are based on those two concepts.

It is possible to construct rational and irrational numbers using straightedge and a compass as seen in chapter II. In 1913 the Indian mathematical Genius, Ramanujam gave a geometrical construction for 355/113 =π. Today with all our accumulated skill in exact measurements. it is a noteworthy feature that lines driven through a mountain meet and make a tunnel.In the earlier classes, we have learnt the construction of angles and triangles with the given measurements.

In this chapter we are going to learn to construct Centroid, Orthocentre, Circumcentre and Incentre of a triangle by using concurrent lines.

4.7.1 Construction of the Centroid of a Triangle

Centroid

The point of concurrency of the medians of a triangle is called the centroid of the triangle and is usually denoted by G.

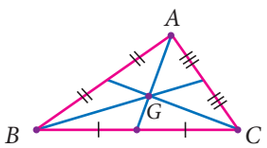


Fig. 4.84



Activity 8

Objective To find the mid-point of a line segment using paper folding

Procedure Make a line segment on a paper by folding it and name it PQ. Fold the line segment PQ in such a way that P falls on Q and mark the point of intersection of the line segment and the crease formed by folding the paper as M. M is the midpoint of PQ.

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